MILITARY SPECIFICATION SHEET

NUT, PLAIN AND SELF-LOCKING, CLINCH, FLOATING, METRIC (SELF CLINCHING, KNURLED COLLAR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nut described herein shall consist of this specification and the latest issue of specification DOD-N-70333.

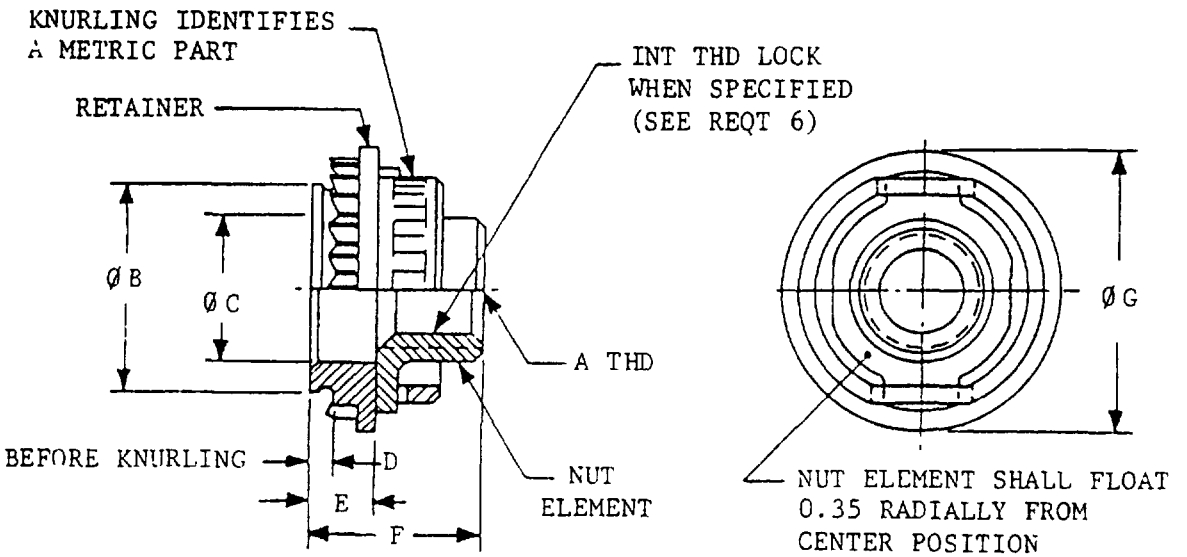


FIGURE 1. Dimensions and configuration.

NOTES:
1. Unless otherwise specified, dimensions are in millimeters and shall apply after plating and before the addition of solid film lubricant.

TABLE I. Part dimensions and dash numbers.

DASH NUMBER		A THD (SEE REQT 5)	ØB	ØC	D	E	F	ØG	MASS Kg/1000 PCS APPROX REF	
ALLOY STEEL	CRES A286		0 -0.15	+0.20 0	MAX	±0.15	±0.45	±0.15	ALLOY STEEL	CRES A286
1A	1C	MJ3x0.5-4H6H	5.98	3.75	0.95	2.05	5.00	8.20	0.740	0.750
2A	2C	MJ4x0.7-4H6H	7.98	5.75	0.95	2.56	6.00	10.60	1.475	1.495
3A	3C	MJ5x0.8-4H6H	7.98	5.75	0.95	2.56	6.50	10.60	1.520	1.540
4A	4C	MJ6x1-4H5H	9.48	6.75	1.30	3.62	8.70	12.70	2.955	2.990

FSC 5310

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REQUIREMENTS:

1. Material: Alloy steel nut element and retainer shall be grade 8740 (UNS G87400) conforming to AMS 6322 or grade 4140 (UNS G41400) conforming to AMS 6382. Optional nut element shall be grade 1050 (UNS G10500) conforming to AMS 5085. Corrosion-resistant steel nut element and retainer shall be type A286 (UNS S66286) conforming to AMS 5731 or AMS 5734. Optional nut element shall be type A286 (UNS S66286) conforming to AMS 5525.

2. Protective finish and surface treatment: Alloy steel nuts (UNS G87400) or (UNS G41400) shall be cadmium plated; in addition, self-locking alloy steel nuts shall also have a solid film lubricant coating. Corrosion-resistant steel nuts (UNS S66286) shall be cleaned, descaled and passivated. Corrosion-resistant steel (UNS S66286) self-locking nuts shall have the nut element silver plated 5 micrometers (µm) thick plus a solid film lubricant coating.

3. Surface texture: Machined surfaces shall be 3.2 µm except knurling.

4. Hardness: Alloy steel nut element (UNS G87400, G41400 or G10500) shall be 35 to 42 HRC. Alloy steel nut retainer (UNS G87400 or G41400) shall be 45 to 50 HRC. Corrosion-resistant steel nut (UNS S66286) shall be 34 HRC minimum.

5. Threads: Threads shall be in accordance with table I and general specification. Thread dimensions shall apply before application of solid film lubricant where applicable.

6. Military part number: Consists of the letter D, plus the basic number of the specification sheet and a dash number taken from table I. Add suffix letter L for self-locking nut. No letter is required for plain nut.

Example of Military part number:

D70333/2-	2A	L
Basic number of specification sheet	Dash number	Self-locking (omit for plain nut)

Explanation of part number example:

D70333/2-2AL = Nut, Self-locking, Clinch, Floating, Metric, MJ4x0.7-4H6H size thread, pitch and tolerance class, alloy steel cadmium plated and solid film lubricant coating.

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7. Resistance to push-out and torque-out: Push-out and torque-out values shall be as specified in table II herein when nut is properly installed in aluminum, type 6061-T6 (UNS A96061) in accordance with QQ-A-250/11, Temper T6.

TABLE II. Push-out and torque-out values.

INT THD SIZE	PUSH-OUT MIN N	TORQUE-OUT MIN N·m
MJ3x0.5	500	2.6
MJ4x0.7	520	4.5
MJ5x0.8	860	6.7
MJ6x1	1200	9.6

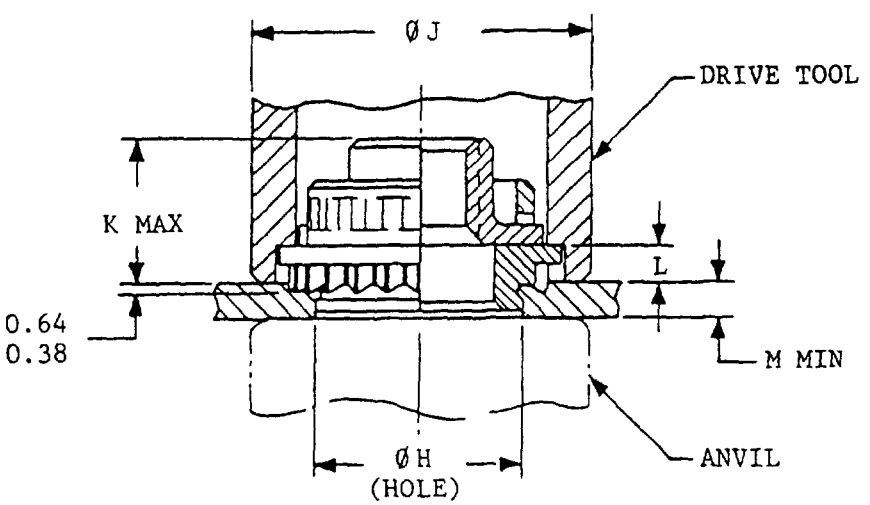


FIGURE 2. Hole preparation and installation.

TABLE III. Installation data in parent material.

DASH NUMBER		ØH	ØJ	K*	L	M
ALLOY STEEL	CRES A286	+0.1 0	REF	MAX	0 -0.2	MIN
1A	1C	6.0	14.3	4.4	1.1	1.2
2A	2C	8.0	19.0	5.4	1.6	1.2
3A	3C	8.0	19.0	5.9	1.6	1.2
4A	4C	9.5	20.6	7.7	2.2	1.8

* K dimension is for design clearance information only. Do not use as installation data.

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Installation procedures.

a. Drill or punch hole (Corrosion-resistant steel shall be drilled).

a.1 Do not chamfer, break or deburr the hole on the side of the material where the nut will be installed.

a.2 In punched holes the nut shall be installed on the side where the punch exits.

b. Locate pilot squarely in hole.

c. Using a drive tool and anvil (as shown in figure 2) apply sufficient force to cause a 0.38-0.64 mm penetration into the structure for a full 360°. Proper penetration is accomplished when shoulder of tool bottoms on surface of parent material. Do not overdrive the nut.

d. Install screw or bolt from pilot side of nut.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Review activities:
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Air Force - 82
NSA - NS
DLA - IS

User activities:
Navy - MC

Preparing activity:
Army - AR

(Project 5310-1327)

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1. DOCUMENT NUMBER	2 DOCUMENT TITLE
DOD-N-70333/2	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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5 PROBLEM AREAS

a. Paragraph Number and Wording.

b Recommended Wording:

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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